

ECHS FRAUD ANALYTICS

FA-10: Time-Series Surge Analytics

Financial Impact & Fraud Report

CLASSIFICATION
RESTRICTED

PERIOD
Jan 2021–Mar 2026

CLAIMS ANALYSED
Over 2.6Cr

HOSPITALS ANALYSED
4,091

IIT KANPUR — Data Analytics & Fraud Intelligence Division

12 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

Executive Summary

The following executive summary details the exact scope and financial exposure identified within the ECHS claims database during the specified timeframe (Jan 2021 - Mar 2026). This module focuses on temporal fraud patterns where hospitals artificially dump or surge billing outside of natural organic growth.

TOTAL CLAIMS ANALYSED 2.6M+ Records evaluated	HOSPITALS ANALYSED 4,091 Empanelled facilities	TOTAL EXPOSURE ₹1,342 Cr Identified risk amount	PATTERNS EXTRACTED 4 Time-series models
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Time-Series Fraud Patterns

Pattern	Description
Pattern 1: End-of-Quarter Surge	Hospitals submitting ≥ 66% of quarterly claims in the final month.
Pattern 2: End-of-Month Dump	Hospitals submitting ≥ 75% of monthly claims in the last week.
Pattern 3: Abnormal Monthly Spikes	Hospitals submitting ≥ 3x their historical median claims in a single month.
Pattern 4: Abnormal YoY Spikes	Hospitals submitting ≥ 3x total claims compared to the previous year.

PATTERN 1

End-of-Quarter Surge

Description: This pattern identifies End-of-Quarter Surges, specifically highlighting instances where 66% or more of a hospital's total quarterly claims were submitted in the last month of that respective quarter.

Table 1.1 — Top Hospitals with EOQ Surges

Hospital Name	Year	Quarter	Quarterly Claims	Month 3 Claims	EOQ Claim %	Quarterly Billed	Exposure
FORTIS HOSPITAL LTD - JAIPUR	2025	1	2,596	1,778	68.49%	₹ 94,565,265	₹ 44,672,603.55
SMSG MULTISPECIALITY HOSPITAL	2025	2	246	172	69.92%	₹ 37,829,018	₹ 23,209,092.18
R P ARORA MEDICITY	2023	4	412	378	91.75%	₹ 35,740,258	₹ 21,809,210.95
KONARC MULTISPECIALITY HOSPITAL	2023	4	210	147	70.00%	₹ 29,416,063	₹ 18,620,120.03
TATA MEMORIAL HOSPITAL, PAREL	2023	2	343	232	67.64%	₹ 26,158,171	₹ 17,553,676.00
Basavatarakam INDO - American Cancer	2024	4	321	216	67.29%	₹ 31,338,986	₹ 14,456,712.25
R J SUPER SPECIALITY HOSPITAL	2023	4	235	165	70.21%	₹ 26,705,703	₹ 13,947,995.59
POSITRON HOSPITAL	2023	1	514	455	88.52%	₹ 17,312,435	₹ 13,154,777.44
RPS HOSPITAL MATERNITY & TRAUMA	2023	1	158	108	68.35%	₹ 24,108,731	₹ 12,967,567.50
MEHTA MULTISPECIALITY HOSPITAL	2023	4	323	240	74.30%	₹ 21,124,120	₹ 12,379,724.76

Key Findings

- Filing massive volumes in the final month of the quarter suggests delayed batching or deliberate obfuscation of claim details to rush past the audit cycle.

PATTERN 2

End-of-Month Dump

Description: This pattern identifies the End-of-Month Dump, specifically isolating hospitals where 75% or more of the claims were submitted in the last week of that month.

Table 2.1 — Top Hospitals with EOM Dumps

Hospital Name	Billing Month	Total Claims	Last Week Claims	Dump %	Total Billed	Exposure
MAX HEALTH CARE INSTITUTE LTD - DEHRADUN	2022-04	483	423	87.58%	₹ 9,167,027	₹ 6,600,890.89
NORTH BENGAL NEURO RESEARCH CENTRE	2025-03	242	189	78.10%	₹ 8,936,539	₹ 3,409,132.79
Secunderabad Main	2023-08	2,059	1,595	77.46%	₹ 12,224,564	₹ 3,113,483.00
PARAS HEALTHCARE PVT LTD - SUSHANT LOK	2024-06	231	187	80.95%	₹ 12,061,182	₹ 2,186,135.20
AMRITA HOSPITAL	2025-10	481	394	81.91%	₹ 5,839,736	₹ 1,999,051.67
GUPTA HOSPITAL & RESEARCH CENTRE	2021-02	136	110	80.88%	₹ 3,453,471	₹ 1,726,139.80
Gurgaon (Sohana Rd)	2023-04	431	399	92.58%	₹ 2,381,748	₹ 1,698,580.00
PRAKASH HOSPITAL P LTD - Noida	2024-11	119	108	90.76%	₹ 3,801,479	₹ 1,649,365.27
RUBAN MEMORIAL HOSPITAL RATAN STONE	2021-09	309	245	79.29%	₹ 2,964,838	₹ 1,448,224.93
GIRIDHAR EYE INSTITUTE PRIVATE LIMITED	2023-02	240	194	80.83%	₹ 1,714,111	₹ 1,209,227.26

Key Findings

- Claim submission should logically follow patient discharge naturally throughout the month. Extreme week-4 concentration indicates either systematic withholding of records or fabricated end-of-cycle generation.

PATTERN 3

Abnormal Monthly Spikes

Description: This pattern detects Abnormal Monthly Spikes, specifically showing only hospitals which have submitted claims over triple (3x) their historical median for a single month.

Table 3.1 — High Multiplier Spikes (Top 10)

Hospital Name	Spike Month	Spike Claims	Historical Median	Multiplier	Amount Billed	Exposure
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-06	543	37	14.68x	₹ 145,246,379	₹ 92,769,595.72
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-03	562	37	15.19x	₹ 138,167,621	₹ 80,469,302.29
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-07	540	37	14.59x	₹ 123,476,340	₹ 80,096,717.14
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-05	405	37	10.95x	₹ 110,775,585	₹ 72,125,497.64
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-09	541	37	14.62x	₹ 111,910,940	₹ 69,782,718.44
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-02	426	37	11.51x	₹ 109,512,368	₹ 62,209,102.78
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2025-10	579	37	15.65x	₹ 109,990,310	₹ 61,754,402.45
DR S P YADAV HOSPITAL	2025-10	495	67.5	7.33x	₹ 97,232,915	₹ 57,552,450.54
DESUN HOSPITAL - KOLKATA	2025-11	1,158	336	3.45x	₹ 90,740,588	₹ 57,224,333.01
PARAS HMRI HOSPITAL - PATNA	2025-09	5,603	1,790	3.13x	₹ 70,242,543	₹ 55,996,682.74

Key Findings

- A hospital billing 15x its normal median in a single month is almost exclusively linked to a sudden change in management strategy towards fraudulent upcoding or agent onboarding.

PATTERN 4

Abnormal YoY Spikes

Description: This pattern identifies Abnormal YoY Spikes, focusing on hospitals which submitted triple (3x) or more claims than the previous year.

Table 4.1 — Top Hospitals with 3x+ YoY Spikes

Hospital Name	Year -4	Year -3	Year -2	Prev Year	Spike Year	YoY Growth	Amount Billed	Exposure
STAR HOSPITAL (A UNIT OF OM MEDICENTRE)	2021: 349	2022: 280	2023: 397	2024: 1646	2025: 5673	244.65%	₹ 1,349,741,946	₹ 762,182,272.41
GRECIAN SUPER - SPECIALITY HOSPITAL	2018: 11584	2019: 8284	2020: 156	2024: 1951	2025: 5857	200.21%	₹ 475,279,739	₹ 371,854,007.04
N/A (Derived Facility)	2018: 4239	2019: 3427	2020: 1436	2021: 969	2022: 3318	242.41%	₹ 439,415,642	₹ 365,667,913.90
N/A (Derived Facility)	2020: 20045	2021: 26992	2022: 788	2024: 223	2025: 15539	6868.16%	₹ 441,055,046	₹ 364,981,028.46
N/A (Derived Facility)	No Data	2021: 165	2022: 240	2023: 536	2024: 1896	253.73%	₹ 361,889,554	₹ 229,535,327.46
N/A (Derived Facility)	2019: 2065	2020: 3327	2021: 5482	2022: 8087	2023: 26026	221.83%	₹ 274,629,397	₹ 223,809,000.62
N/A (Derived Facility)	2020: 476	2021: 349	2022: 280	2023: 397	2024: 1646	314.61%	₹ 345,320,316	₹ 217,271,595.04
N/A (Derived Facility)	2019: 636	2020: 115	2021: 54	2022: 407	2023: 3165	677.64%	₹ 257,944,153	₹ 212,565,775.63

Key Findings

- A Year-over-Year growth exceeding 200-300% without corresponding capacity increases strongly flags hospitals onboarding aggressive marketing/billing agents.